

1. What is Operator Precedence?

Operator precedence tells C++ **which operator should be performed first** when an expression contains more than one operator.

Example

```
int result = 5 + 3 * 2;
```

Multiplication `*` has higher precedence than addition `+`.

So:

```
5 + (3 * 2)
= 5 + 6
= 11
```

Answer: 11

2. Basic Rule

Remember:

Higher precedence → performed first

For example:

```
10 - 4 / 2
```

`/` has higher precedence than `-`.

```
10 - (4 / 2)
= 10 - 2
= 8
```

3. Common Operator Precedence in C++

For beginners, remember this order:

Priority	Operators	Meaning
1	<code>()</code>	Parentheses
2	<code>++ --</code>	Increment / Decrement

Priority	Operators	Meaning
3	* / %	Multiplication, Division, Modulus
4	+ -	Addition, Subtraction
5	< <= > >=	Relational
6	== !=	Equality
7	&&	Logical AND
8	`	Logical OR
9	= += -= *= /=	Assignment

Easy way to remember

```
( )  
↓  
++ --  
↓  
* / %  
↓  
+ -  
↓  
< <= > >=  
↓  
== !=  
↓  
&&  
↓  
||  
↓  
=
```

4. Parentheses Have the Highest Priority

Use parentheses () when you want something to be calculated first.

```
int result = (5 + 3) * 2;
```

First:

```
5 + 3 = 8
```

Then:

$8 * 2 = 16$

Answer: 16

Compare:

```
int result = 5 + 3 * 2;
```

Answer:

11

But:

```
int result = (5 + 3) * 2;
```

Answer:

16

5. $*$, $/$, and $\%$

These operators have the same precedence.

```
int x = 20 / 5 * 2;
```

When operators have the **same precedence**, C++ generally evaluates them **from left to right**.

```
20 / 5 = 4  
4 * 2 = 8
```

Answer: 8

Another example

```
int x = 17 % 5 + 2;
```

First $\%$:

$$17 \% 5 = 2$$

Then:

$$2 + 2 = 4$$

Answer: 4

6. + and -

+ and - have the same precedence and are evaluated from **left to right**.

```
int x = 10 - 3 + 2;
```

First:

$$10 - 3 = 7$$

Then:

$$7 + 2 = 9$$

Answer: 9

7. Example: Complete Expression

Consider:

```
int result = 5 + 2 * 3;
```

Step 1: *

$$2 * 3 = 6$$

Step 2: +

$$5 + 6 = 11$$

Therefore:

```
result = 11
```

8. Example with Parentheses

```
int result = (5 + 2) * 3;
```

Step 1

```
5 + 2 = 7
```

Step 2

```
7 * 3 = 21
```

Therefore:

```
result = 21
```

9. Important Rule: Left-to-Right

When two operators have the **same precedence**, they are generally evaluated from **left to right**.

Example:

```
int result = 20 / 5 * 2;
```

```
20 / 5 = 4  
4 * 2 = 8
```

So:

```
Answer = 8
```

Not:

$$20 / (5 * 2) = 2$$

10. Best Practic

When an expression becomes difficult to understand, **use parentheses**.

Instead of:

```
int result = a + b * c - d / e;
```

Think of it as:

```
int result = a + (b * c) - (d / e);
```

Parentheses make the order clear and reduce mistakes.

Quick Revision

Remember this order:

Parentheses → **Increment/Decrement** → **Multiply/Divide/Modulus** → **Add/Subtract** → **Relational** → **Equality** → **AND** → **OR** → **Assignment**

```
()  
++ --  
* / %  
+ -  
< <= > >=  
== !=  
&&  
||  
=
```

Golden Rule ★

First check parentheses, then higher-precedence operators, and when operators have equal precedence, follow their associativity (commonly left-to-right for arithmetic operators).